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Prediction of Drug-Disease Associations for Drug Repositioning Through Drug-miRNA-Disease Heterogeneous Network

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ABSTRACT Drug repositioning, which refers to the identification of new clinical indications for existing drugs, has become an important strategy for drug discovery. The most recent studies in pharmacogenomics have demonstrated that drugs can target microRNAs (miRNAs) and regulate their expression levels. Given the intriguing fact that the inappropriate expression of miRNAs is related to many kinds of human diseases, developing small-molecule drugs to target specific miRNAs and modulate their activities would be a promising approach to disease treatment, which offers an innovative insight for drug repositioning. In this paper, we proposed a miRNA-based computational method HNBI to infer novel drug-disease associations for drug repositioning. Similarity measurements and experimentally supported association information were first integrated to construct a three-layer drug-miRNA-disease heterogeneous network. Our method then updated the strength of weight between unlinked drug-miRNA, miRNA-disease, and drug-disease pairs iteratively till stabilized. Based on information flow on the heterogeneous network, the final weight of drug-disease associations was received by summarizing the values of paths connecting the two types of nodes. We prioritized the potential drug-disease associations according to the new weight. When applied to the collected data set for cross-validation experiments, our method showed superior performance in drug-disease association predictions compared with two state-of-the-art methods. Furthermore, our method HNBI incorporated information of target miRNAs to understand the mechanisms of action of drugs and the molecular mechanisms of diseases. A case study on the drug *Terazosin* indicated that some predicted indications with high ranks were supported by the recent literature, which further illustrated the practical usefulness of our method. Finally, comprehensive predictions of associations between drugs and diseases were released for future drug repositioning studies.

INDEX TERMS Drug repositioning, target miRNAs, heterogeneous network, graph inference.

I. INTRODUCTION

Over the past decades, spending on research and development (R&D) in the pharmaceutical industry has increased enormously. According to statistical analysis, the total worldwide pharmaceutical R&D expenditures have grown to some 157 billion U.S. dollars in 2016 compared with less than 10 billion U.S. dollars in 1990 and this figure will increase to more than 180 billion U.S. dollars in 2022 (<https://www.statista.com/statistics/309466/global-r-and-d-expenditure-for-pharmaceuticals/>). Despite such huge growth in spending, productivity in pharmaceutical companies has not kept pace. Only 46 novel drugs were approved by U.S.

Food and Drug Administration (FDA) in 2017 compared to 39 new drugs endorsed in 2012 (<https://www.fda.gov/Drugs/DevelopmentApprovalProcess/DrugInnovation/default.htm>).

In the meanwhile, there exists increasing regulation and worldwide pressure on prices for pharmaceutical companies [1]. In order to tackle these challenges, many drug developers are becoming more and more creative in identifying new uses for existing drugs, which is also known as drug repositioning [2]. One of the most renowned repositioning stories is Pfizer's *Viagra*, which was originally developed for cardiovascular disease treatment and was later repurposed as an effective drug in mediating erections [2].

The main challenge lied in drug repositioning is to select the most therapeutic effects for a drug of interest for carrying out clinically prospective tests. Traditional repositioning efforts include blind screening [3]–[5] and serial testing of animal models [6]. With the accumulation of biomedical data, computational approaches provide an effective alternative to predicting the most promising drug-disease associations for new indication test for existing drugs.

For instance, Chiang and Butte [7] implemented a guilt-by-association method for novel drug use suggestions based on shared treatment profiles from disease pairs. The rationale for this methodology was that if two diseases shared some similar therapies, then other drugs that were therapeutic for only one of the two diseases might also be used for the other. Based on the assumption that similar drugs were indicated for similar diseases, Gottlieb *et al.* [8] utilized multiple features to calculate drug-drug and disease-disease similarity measures and designed a novel classification algorithm, PREDICT, for comprehensively inferring drug indications. Luo *et al.* [9] improved the similarity measures used in Reference [8] and developed a random-walk-based method MBiRW to predict potential drug-disease associations. Yang and Agarwal [10] exploited clinical side-effects (SEs) as human phenotypic profiles for drugs and built Naive Bayes models to predict indications for chemical compounds. Chen *et al.* [11] formulated the problem of drug repositioning as recommending preferable diseases for drugs. Topology of the drug-disease bipartite network was applied to prioritize potential diseases for drugs of interest.

Satisfied prediction performance could be received from the methods listed above. Nevertheless, these methods offered no molecular hints to help design biomedical experiments to test and confirm the results. Therefore, it is of great importance to investigate how drugs and disease phenotypes are associated on the basis of molecular levels.

More recently, studies (see Reference [12] for more details) in pharmacogenomics have discovered that drugs can target microRNAs (miRNAs), which are short (~22 nucleotide) non-coding RNAs, and regulate their expressions. For example, Tilghman *et al.* [13] discovered that the expression profile of *miR-21* in MCF-7 breast cancer cells could be altered by two well-known endocrine disrupting compounds BPA and DDT. Meanwhile, miRNAs function as important regulators of gene expression at the post-transcriptional level and control some essential cellular processes [14]. Therefore, the inappropriate expression levels of miRNAs are closely related to many complex human diseases [15]–[19]. As such, targeting miRNAs with small-molecule drugs provides a promising approach to human disease treatment [20], which gives researchers a new perspective for drug repositioning.

With this understanding, developing computational methods to predict potential drug-disease associations through the target miRNAs is of critical need. Chen *et al.* [21] proposed a miRNA-based method to infer drug-disease associations under the assumption that drugs would associate

with diseases when they shared some significant miRNAs. In this method, experimentally supported drug-miRNA associations and miRNA-disease associations were first integrated. Hypergeometric tests were then applied and *p* values received were used to classify drug-disease pairs. This is the first computational effort made to predicting drug-disease associations under the perspective of miRNAs. As only topology of networks was used, this method was not feasible for new drugs whose drug-miRNA association information was not available. Recently, Chen *et al.* [22] developed a bi-random-walk-based method miRDDCR to discover drug-disease causal relationships through the molecular level of miRNAs by integrating known drug-miRNA associations, miRNA-disease associations and similarity measures. miRDDCR could be implemented to predict associated diseases for new drugs whose target miRNA information could not be obtained. Compared with the method in Reference [21], miRDDCR showed better predicting performance. However, the overall prediction ability of miRDDCR still needs to be improved.

In this manuscript, we proposed a heterogeneous-network-based inference method, HNBI, for novel drug indication predictions based on an intuitive interpretation of information flow on the drug-miRNA-disease heterogeneous graph. More specifically, a three-layer drug-miRNA-disease network, which incorporated experimentally confirmed drug-miRNA, miRNA-disease and drug-disease associations as well as similarity measures, was first constructed. The algorithm HNBI then iteratively updated the strength of weight between unlinked drug-miRNA, miRNA-disease and drug-disease pairs. When eventually converged, we combined the weights of the paths connecting the drug nodes and the disease nodes, and ranked the drug-disease associations according to the new weights. We tested the prediction ability of HNBI on large-scale cross validations. Experimental results showed that our method was superior in prioritizing drug indications when compared with two state-of-the-art methods. A case study on the drug *Terazosin* suggested some high-ranked predictions were verified by recent literature, which further indicated the practical usefulness of our method. After confirming the prediction ability of our method, we finally conducted a comprehensive drug-disease association prediction for drug repositioning.

II. MATERIALS AND METHODS

A. DATASETS

The information of drugs, miRNAs and diseases used in our manuscript were downloaded from different data sources. All pairwise (drug-drug, miRNA-miRNA and disease-disease) similarity measurements were first calculated. Experimentally confirmed drug-miRNA, miRNA-disease and drug-disease associations were then selected from existing databases. Finally, a drug-miRNA-disease heterogeneous network (Figure 1) was constructed based on the above information. Here below we provided a brief description of the datasets.

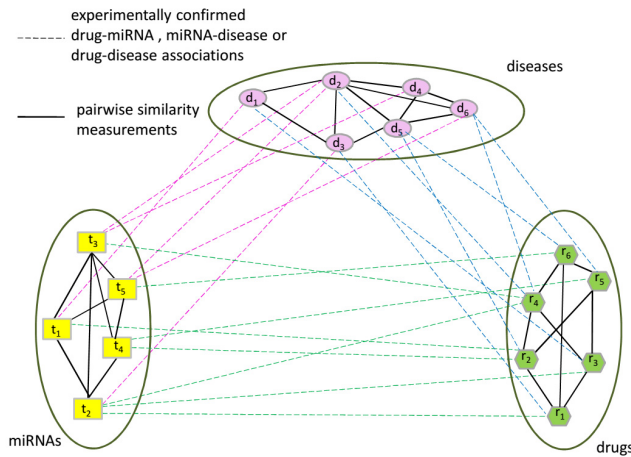


FIGURE 1. A drug-miRNA-disease heterogeneous network was constructed based on drug-drug similarities, miRNA-miRNA similarities, disease-disease similarities, drug-miRNA associations, miRNA-disease associations and drug-disease associations.

Drug-drug similarities Small molecular drug compounds were first compiled from DrugBank [23], PubChem [24] and SM2miR [12]. Redundant records across the three databases were deleted. Drug-drug similarities were calculated based on their chemical structures [25], functional consistency [26] and side effects [8]. Similar to Reference [27], the integrated similarity value was received by averaging the results based on the three features.

miRNA-miRNA similarities We obtained miRNAs from SM2miR [12], HMDD [28], miR2Disease [29] and PhenomiR [30]. Functional consistency [26] of target gene sets was applied to determine miRNA-miRNA similarities.

Disease-disease similarities We obtained the diseases and their pairwise similarity measurements from Reference [31]. The similarity was calculated by incorporating disease semantic similarity and disease functional similarity.

Drug-miRNA associations Experimentally confirmed drug-miRNA associations were collected from SM2miR [12]. We only selected the records constrained within the species of *Homo sapiens* in the database.

miRNA-disease associations Experimentally validated miRNA-disease associations were obtained from HMDD [28].

Drug-disease associations Experimentally verified associations between drugs and diseases were extracted from Comparative Toxicogenomics Database (CTD) [32], DrugBank [23] and Therapeutic Targets Database (TTD) [33]. For the three types of associations, duplicated records in the databases were retained only once for consideration in our datasets.

Finally, we received 831 drugs, 540 miRNAs, 341 diseases, 630 drug-miRNA associations, 6082 miRNA-disease associations and 13,490 drug-disease associations.

B. METHOD DESCRIPTION

The drug-miRNA-disease heterogeneous network consists of three types of nodes, i.e. drug nodes, miRNA nodes and disease nodes, and six types of edges, i.e. drug-drug edges, miRNA-miRNA edges, disease-disease edges, drug-miRNA edges, miRNA-disease edges and drug-disease edges. Let $R = \{r_1, r_2, \dots, r_m\}$ denote the drug nodes, $T = \{t_1, t_2, \dots, t_n\}$ represent the miRNA nodes and $D = \{d_1, d_2, \dots, d_k\}$ indicate the disease nodes. Let E_{rr} , E_{tt} , and E_{dd} represent the intra-edges between drugs, miRNAs and diseases, and W_{rr} , W_{tt} , and W_{dd} indicate the weights on the three kinds of edges. We assigned the pairwise similarity values to W_{rr} , W_{tt} , and W_{dd} . Let E_{rt} , E_{td} and E_{rd} denote known drug-miRNA, miRNA-disease and drug-disease associations, and W_{rt} , W_{td} and W_{rd} indicate the weights on the three types of edges. For W_{rt} , W_{td} and W_{rd} , we set the value to be 1 when there existed an association, otherwise 0.

Our current knowledge of drug-miRNA associations, miRNA-disease associations and drug-disease associations is far from complete, which indicates that the original heterogeneous network can be considered as an incomplete graph with missing edges between the three types of nodes. The drug-disease association prediction problem can then be transformed into a missing edge prediction problem on the heterogeneous network. Our objective is to capture hidden edges between drug nodes and disease nodes.

Inspired by the successful work of Reference [34], for each initially unconnected drug-miRNA pair, its new weight can be calculated according to the following equation by combining the similarity measurements and existing associations,

$$w(r, t)_{new} = \sum_{r_i \in R} \sum_{t_j \in T} w(r, r_i) \times w(r_i, t_j) \times w(t_j, t) \quad (1)$$

Where r is a drug and t is a miRNA. r_i and t_j are the neighbors of r and t , respectively. Once a new weight between the drug r and the miRNA t is established, more associations between drugs and miRNAs can then be generated iteratively according to the following equation

$$W_{rt}^{p+1} = W_{rr} \times W_{rt}^p \times W_{tt} \quad (2)$$

We consider the initially existing associations between drugs and miRNAs differently from the inferred drug-miRNA associations. Equation (2) can be transformed as

$$W_{rt}^{p+1} = \alpha W_{rr} \times W_{rt}^p \times W_{tt} + (1 - \alpha) W_{rt}^0 \quad (3)$$

Where α is a decay factor W_{rt}^0 and represents the initial values of associations between drugs and target miRNAs. When equation (3) reaches a steady state, novel drug-miRNA associations can be predicted according to the new values of W_{rt} .

Similarly, to predict new miRNA-disease associations and drug-disease associations, we define W_{td}^{new} and W_{rd}^{new} as

$$W_{td}^{new} = W_{td}^{q+1} = \beta W_{tt} \times W_{td}^q \times W_{dd} + (1 - \beta) W_{td}^0 \quad (4)$$

$$W_{rd}^{new} = W_{rd}^{s+1} = \gamma W_{rr} \times W_{rd}^s \times W_{dd} + (1 - \gamma) W_{rd}^0 \quad (5)$$

Intuitively, based on the information flow on the three-layer heterogeneous network, drug-disease association predictions can be received by combining two separate chains, i.e. drug nodes $\rightarrow$ miRNA nodes $\rightarrow$ disease nodes and drug nodes $\rightarrow$ disease nodes. Therefore, predictions of associations between drugs and diseases can be finally achieved by summarizing the results of the two chains as

$$W_{rd}^{final} = W_{rt}^{new} \times W_{td}^{new} + W_{rd}^{new} \quad (6)$$

For each drug, its associated diseases will be ranked according to the value of equation (6).

III. RESULTS

A. EXPERIMENTAL DESIGN AND EVALUATION METRICS

In order to comprehensively evaluate the performance of our method in predicting drug-disease associations, we adopted a leave-one-out cross-validation (LOOCV) strategy in our experiments. For each drug in our dataset, it was considered as a test drug once and its association information was deleted. The remaining 830 drugs were taken as the training dataset. The predicted indications for the test drug were ranked according to the final result received by equation (6). We treated the 13,490 known drug-disease associations as positive instances. For each specific ranking threshold, if the weight of an inferred drug-disease association was above the threshold, it was regarded as a true positive. Otherwise, it was regarded as a false positive. True positive rate (TPR) and false positive rate (FPR) were calculated by varying thresholds to construct ROC curves. We used the area under the curve (AUC) value to represent the prediction ability of our method. When AUC value equals 1, we can declare that the method has perfect prediction performance; An AUC value of 0.5 indicates random prediction performance.

To further analyze the practical applications of our algorithm, we conducted a case study and selected the top predicted drug-disease associations for confirmation. Moreover, we used the intermediate results to discover potential drug-miRNA associations and miRNA-disease associations for revealing the mechanisms of action of drugs and for understanding the molecular mechanisms of diseases.

B. COMPARISON WITH OTHER METHODS

Previously, two computational methods [21], [22] were proposed to predict drug-disease associations based on target miRNAs. To evaluate the performance of our approach HNBI, we chose to compare it with the two state-of-the-art methods.

The method developed in Reference [21] applied hypergeometric tests to determine potential drug-disease associations. As this method used only known drug-miRNA associations and miRNA-disease associations, it could not be used to infer drug-disease associations for those drugs whose target miRNAs were not available. The method HNBI in our manuscript and the method miRDDCR in Reference [22] could overcome this drawback.

In order to have comparable results, we performed leave-one-drug-out cross-validation experiments on the benchmark dataset for both methods miRDDCR [22] and HNBI. The experimental procedure considered each drug as a test drug and its association information was removed from the dataset. The predicted drug-disease associations were ranked and we calculated TPRs and FPRs to plot ROC curve. As there existed 201 drugs whose experimentally confirmed drug-disease associations were not available in the gold standard dataset, we calculated the AUC values for the rest 630 drugs. A comparison of the distribution of AUC values for the two methods could be seen in Figure 2. For HNBI, we received an average AUC value of 0.781 compared with an average AUC value of 0.717 for miRDDCR. Moreover, we received higher AUC values than 0.7 for 469 drugs when HNBI was implemented. While there were only 376 drugs whose AUC values were larger than 0.7 when miRDDCR was applied. The experimental results showed that HNBI outperformed the other two methods significantly.

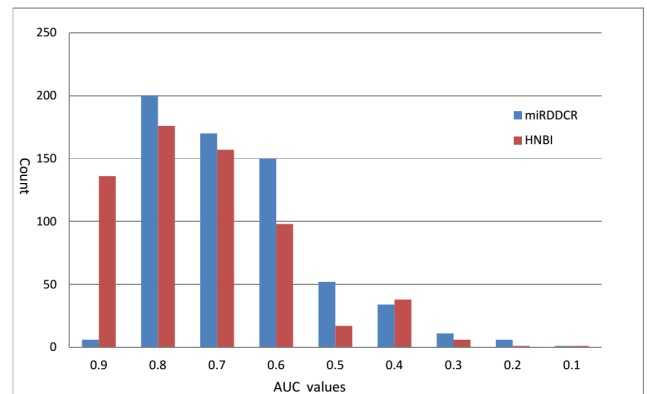


FIGURE 2. A comparison of distribution of AUC values received by the methods miRDDCR and HNBI when leave-one-drug-out cross-validations were implemented.

C. PREDICTIONS OF DRUG-miRNA ASSOCIATIONS AND miRNA-DISEASE ASSOCIATIONS

It was also worth pointing out that our method HNBI was able to infer drug-miRNA and miRNA-disease associations by using equation (3) and (4) in the Materials and Methods section, respectively. As miRNAs were critical biomarkers in drug repositioning in this study and experimentally confirmed drug-miRNA associations and miRNA-disease associations were limited, computational predictions of such associations can provide significant help. We first conducted leave-one-out cross validations to test the prediction performance. Experimental results showed that when leave-one-drug-out cross validations were implemented, an average AUC value of 0.716 was received in drug-miRNA association predictions; when leave-one-miRNA-out cross validations were realized, an average AUC value of 0.780 was received in miRNA-disease association predictions. These results indicated that reliable drug-miRNA association and miRNA-disease association prediction results could be achieved by our method. We then applied equation (3) and (4)

to the gold standard dataset to infer drug-miRNA and miRNA-disease associations. Top 20 predicted results for each type of associations were reported in Supplementary material S1 and S2, respectively.

D. A CASE STUDY

In addition to large-scale cross-validation experiments for performance evaluation, we also applied our method HNBI to the collected dataset for practical usage. We selected the drug *Terazosin* (DB01162) for a case study. For this drug, there existed no target miRNA information in our drug-miRNA association dataset. There was only one drug-disease association relating to this drug in our gold standard dataset and we deleted this information. The method HNBI was then applied and we chose to rank the predicted indications for this drug in a descending way. We discovered that *hypertension*, which was the only indication information for the drug in the gold standard dataset, was ranked 27th out of the 341 candidate diseases. Moreover, we took the top 5 prediction results (*cardiovascular diseases*, *neoplasms*, *vascular diseases*, *multiple myeloma* and *liver neoplasms*) for further confirmation. We searched the literature and found that two diseases, *cardiovascular diseases* (ranked 1st) and *vascular diseases* (ranked 3rd) were supported by published papers [35]–[37].

As our method HBNI could automatically infer drug-miRNA associations and miRNA-disease associations by using equation (3) and equation (4), respectively. For *Terazosin*, the top 5 predicted target miRNAs were *hsa-mir-338*, *hsa-mir-200b*, *hsa-mir-204*, *hsa-mir-195* and *hsa-mir-200a*. A study by Courboulin *et al.* [38] demonstrated that the down-regulation of *has-mir-204* (ranked 3rd) correlated with pulmonary arterial hypertension (PAH) severity. Another two studies [39], [40] implied that the concentration of *has-mir-195* (ranked 4th) was associated with myocardial diseases.

All these results showed that the proposed approach HNBI could be very effective in not only predicting drug-disease associations for drug repositioning, but also providing useful information for investigating the mechanisms of action of drugs and for understanding the development and progression of human complex diseases, which we anticipated would greatly provide help to wet lab experimental test.

E. COMPREHENSIVE DRUG-DISEASE ASSOCIATION PREDICTION FOR DRUG REPOSITIONING

In this experimental scenario, we widely inferred drug-disease associations for future drug repositioning usage. We trained HNBI with similarity measurements and all the known association information. We ranked the non-interacting drug-disease pairs with respect to their new weights and selected the top 20 predicted indications for each drug. The full lists of the predicted drug-disease associations could be available from Supplementary material S3 for further studies.

IV. DISCUSSION

Repositioning existing drugs for new indications could benefit both pharmaceutical companies and human beings much. Computational predictions of drug-disease associations are effective ways to provide drugs with the most promising indication candidates for further biomedical tests. Therefore, developing efficient algorithms to infer drug-disease associations is of importance and numerous efforts have been made to this field.

More recently, advances in pharmacogenomics indicated that miRNAs function as critical factors in drug response and disease treatment. In this manuscript, we proposed a computational method HNBI to predict drug-disease associations through a drug-miRNA-disease heterogeneous network. Compared with existing methods, we incorporated more reliable features for drug-disease association predictions. Experimental results showed that HNBI outperformed other two popular methods. A case study on the drug *Terazosin* indicated that the results inferred by our method were supported by existing knowledge. In particular, as a by-product, our method was capable of inferring new drug-miRNA associations and miRNA-disease associations, and the information of target miRNAs gave reasonable explanations to the cause-and-effect mechanisms behind drug-disease associations.

There are two factors contributing to the good performance of our approach. First, experimentally supported chemical and biological datasets were utilized to construct the drug-miRNA-disease heterogeneous network. Second, our algorithm iterately captured the hidden relationships between drug-target miRNA, miRNA-disease and drug-disease for final drug-disease association predictions.

One should notice that the mechanisms of action of drugs are complex. Previous research mainly considered proteins [41]–[43] as drug targets. Some methods have used proteins [44]–[46] or gene expression data [47], [48] to predict novel indications for existing drugs. Combinatorial therapies [49], [50], which aim at reducing drug side effects and improving drug efficacy through drug combinations, indicate another promising strategy for drug repositioning. These studies all made contribution to drug discovery.

As newly discovered targets, our investigation on miRNAs is not completed and many drug-miRNA and miRNA-disease associations have not been detected. We expect that the performance of our method could be improved by integrating more experimentally confirmed association information. Similarity is also a critical factor in determining the prediction power of our method. Incorporating some other properties, like miRNA expression data, may produce more accurate prediction results.

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